No-Trim RIAA Stage: Employs Analog Computer

Mark S. Williamsen

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Abstract

Listening to phonograph records usually involves a de-emphasis filter to recover the original sound. In this article we explore how to implement a state-variable filter for this purpose using the Burr-Brown UAF42 integrated circuit. Analog computing techniques allow us to build a circuit matching the RIAA curve precisely with no binning or trimming of parts.

I recently bought a new turntable with a moving magnet (MM) cartridge, and had in mind to create an RIAA phono preamp to go with it. My idea was to employ the principles of analog computing to build a no-trim RIAA de-emphasis circuit with high precision. One that would work well with purchased components, and achieve the desired accuracy without binning or trimming parts. Which meant turning to state-variable techniques to realize the three time constants prescribed for RIAA equalization [1]. It's easy to buy 0.1% metal film resistors from distributors. A 0.1% gain error would amount to less than 0.01 decibels. But that doesn't solve the problem of how to obtain capacitors with similar precision needed for the time constants. Which led me to consider the UAF42 universal active filter, originally from Burr-Brown and now available from Texas Instruments [2]. The UAF42 implements a classic two-integrator state-variable filter, with internal 0.5% precision capacitors and resistors.

As I started looking at the RIAA de-emphasis transfer function, I recalled some advice given by Don Lancaster in his excellent "Active Filter Cookbook" [3], that it is often easier to develop a higher-order filter by cascading multiple lower-order sections. That allows one to develop and test each section independently. This was the insight that allowed me to proceed quickly, and I would recommend it to anyone reading this.

$$H(s) = \left(\frac{1}{sT_3 + 1}\right)\left(\frac{sT_2 + 1}{sT_1 + 1}\right) \tag{1}$$

So I could do the whole job with two UAF42s, each configured as two first-order sections. Realizing I would need up to 60 decibels of voltage gain at low frequency, I set the gain of each filter section to 20 decibels, and then added a gain stage in front with another 20 decibels. The low-pass filter stage rolls off at 6 decibels per octave with a corner frequency of 2122 hertz. The shelf filter and output stage has a slope of 6 decibels per octave between two corner frequencies, at 50 and 500 hertz. I used the BodeS Java application to visualize

this [4]. Since my focus was now on developing a no-trim RIAA equalizer using the UAF42, I tried several different input stages and finally settled on a simple complementary pair of bipolar junction transistors (BJTs). The complementary pair is non-inverting from input to output, so I was happy to configure both filter sections as inverting. This resulted in an overall non-inverting design.

While the filter stages are self-biasing near ground, the input stage on the other hand required active measures for its output to sit near ground. I fitted a servo amplifier to control the input stage bias point. It was convenient to connect the servo output to an inverting input on the input stage. To close the loop with negative feedback then required a servo input which is the inverse of the gain stage output. And since my filter sections are both inverting, my solution was to take the servo input from the output of the inverting low-pass filter stage.

State variable filters have good properties to begin with. Implementing them with opamps adds more good properties. And then using a UAF42 makes it even better still! First of all, state variable filters are gain-tuned. So if we have to tweak them it can all be done with resistors and potentiometers. Then using opamps means we benefit from arbitrarily high input impedance and open loop gain, and arbitrarily low output impedance. This means we can get precise gains just by using precise resistors. Finally choosing the UAF42 means our integrating capacitors are already inside, calibrated with stray capacitance included.

1 Schematic Diagram

A schematic diagram of the left channel is shown in figure 1. Most of it is duplicated for the right channel with the exception of U_1 which contains two opamps, one for each channel. The first filter section at U_{2A} implements a first order low-pass filter with DC voltage gain A_v and time constant T_3.

$$\begin{aligned} A_v &= -\frac{R_8}{R_7} = -10, \\ T_3 &= R_8 C_3 = 75 \text{ microseconds}. \end{aligned} \tag{2}$$

The second filter section at U_{2B}, U_{2C}, and U_{2D} implements a first order shelf filter with DC voltage gain A_v and time constants T_1 and T_2.

$$\begin{aligned} A_v &= -\frac{R_{11}}{R_{10}} = -10, \\ T_1 &= \frac{R_{11} R_{13} R_{14} C_4}{R_{12} R_{16}} = 3180 \text{ microseconds}, \\ T_2 &= \frac{R_{10} R_{13} R_{14} C_4}{R_{12} R_{15}} = 318 \text{ microseconds}. \end{aligned} \tag{3}$$

The gain stage at Q_1 and Q_2 forms a complementary pair of BJTs with gain controlled primarily by R_5 and R_4. This stage serves several purposes. It has a fairly high input impedance so the load seen by the cartridge is close to the parallel combination of C_1, R_2, and capacitance in the tone arm and connecting

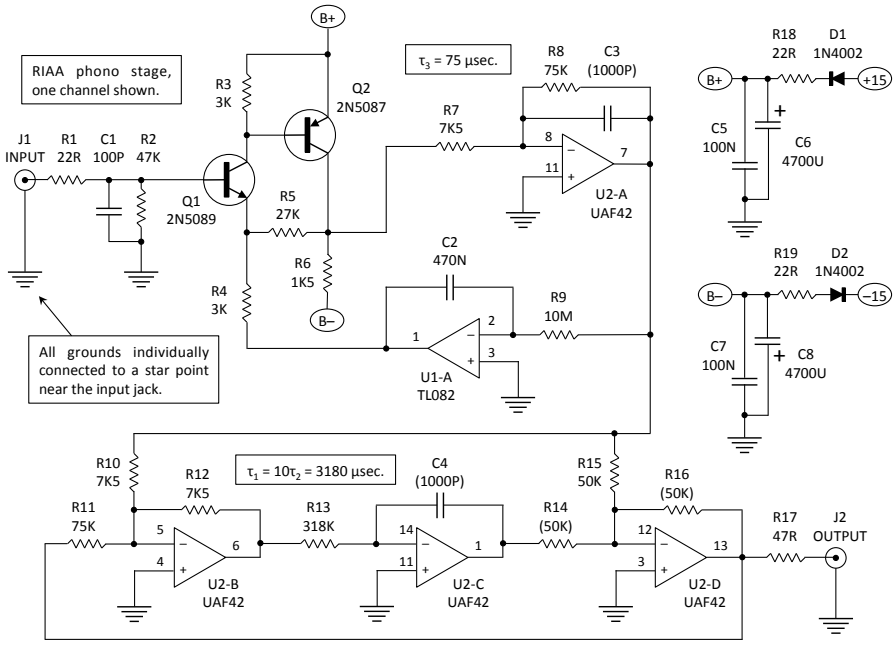


Figure 1: Schematic diagram of the state-variable RIAA stage, one channel shown. Filtered and regulated linear power supply (not shown) enters through D_1 and D_2. Connect U_1 pin 8 to $B+$ and U_1 pin 4 to $B-$. Connect U_2 pin 10 to $B+$ and U_2 pin 9 to $B-$.

cables. The values of C_1 and R_2 can be adjusted to suit the cartridge you are listening to. The input stage has a fairly low output impedance, able to drive the input of the low-pass filter which follows. Finally it has enough gain to get us up off the noise floor, before we start filtering. A simple approximation of gain A_v can be written in the same form we would use for an non-inverting voltage-feedback opamp stage.

$$A_v = 1 + R_5/R_4 = 10 \tag{4}$$

My prototype however produced a different gain which led me to look closer at how the stage works. I concluded that Q_1 functions as an emitter-follower with near-unity voltage gain. The emitter of Q_1 is then a current-summing node which serves as the inverting input to the stage. The current gain β of Q_2 amplifies its base current, which follows the emitter current of Q_1, to produce its collector current which is divided between R_5 and R_6. In other words, the input stage is functioning as a current-feedback amplifier. With this analysis I've written a much better approximation of voltage gain for the stage. The numeric result agrees with my prototype if I assume $\beta \approx 200$.

$$A_v = \frac{1 + \beta(1 + R_5/R_4)}{1 + \beta + R_5/R_6} = 9.13\cdots \tag{5}$$

Taking the limit as β goes to infinity recovers the original approximation in equation 4. The servo amplifier at U_{1A} implements a simple integrator that biases the input stage. This acts so that the low pass filter output at pin 7 of U_{2A} sits at ground plus an offset voltage. It looks like the servo has a time constant of $R_9C_2 = 4.7$ seconds. Recalling however that the input gain stage and low pass filter section have a combined DC gain of nearly 100, the time constant is actually around 0.05 seconds. This results in a high pass corner frequency of $f = 1/(2\pi\tau) \approx 3.2$ hertz.

2 Construction

I started breadboarding with my Heathkit ET-3200 Design Experimenter shown in figure 2. After I gained some experience with the design I moved it over to a pre-printed circuit board from Twin Industries in figure 3, laid out in the same pattern as a breadboard socket. Stray capacitance between adjacent pins is generally less on the circuit board, so overall performance improved slightly after the move. The circuit board doesn't have a ground plane, so I mounted it on the lid of the project box and considered that to be my ground plane.

For simplicity I used a packaged open-frame linear power supply capable of 0.4 amps at ± 15 volts. Actual current draw was about one tenth of that, 40 milliamps. While the preamp as shown has relatively good power supply rejection, the fact that DC gain is nearly 1000 means it is especially sensitive to line frequency ripple and harmonics. I added low pass filters $R_{18}C_6$ and $R_{19}C_8$ on the inbound supply lines with corner frequencies of 1.5 hertz, which was enough to make hum inaudible compared to thermal noise, at the expense of a little headroom. Reverse protection diodes D_1 and D_2 cost a little more headroom. C_5 and C_7 are placed near the UAF42s for power supply decoupling. I placed the power supply in a separate chassis on the floor, to keep stray fields

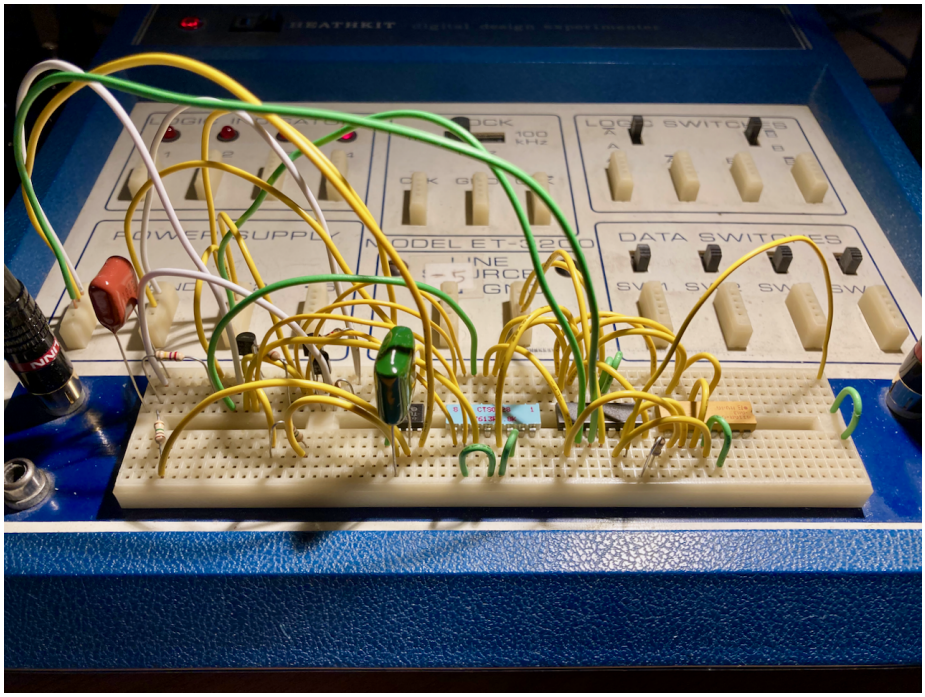


Figure 2: Heathkit ET-3200 breadboard with a UAF42 implementing one channel of RIAA de-emphasis. The UAF42 worked fine with the ± 12 volt power supplies in the ET-3200, but with some loss of headroom. DIP resistor networks on the breadboard were replaced with precision metal film resistors in the final prototype.

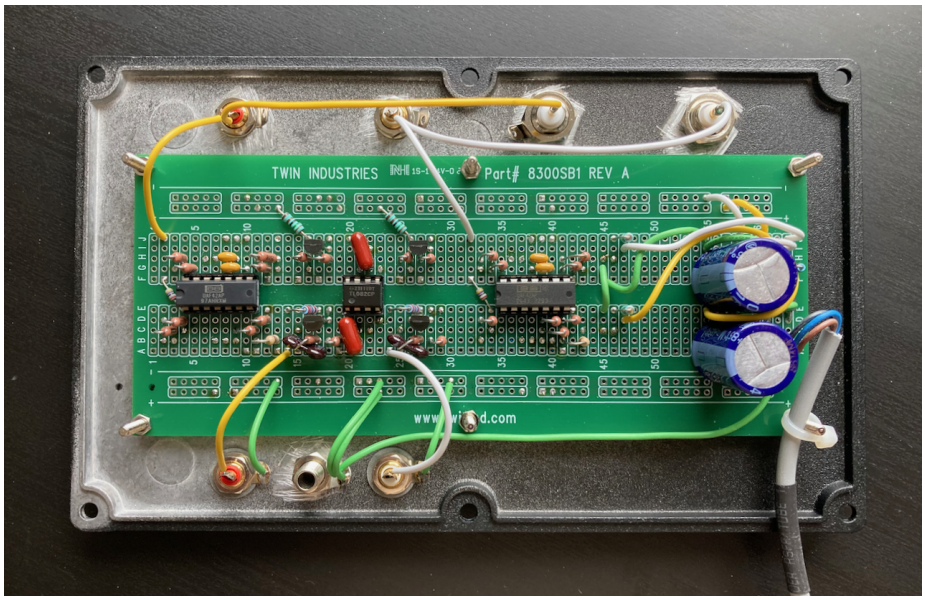


Figure 3: Populated Twin Industries printed circuit board (PCB) mounted inside the lid of a pre-painted Hammond project box. Inputs and ground star point are visible at bottom left, outputs are in the top row. BNC jacks at top right are provided for connection to an oscilloscope.

away from the phono cartridge and preamp. I connected the turntable ground lead to the preamp's metal enclosure at a star point near the input jacks. As an added precaution I mounted the input jacks on fiber shoulder washers so I could ground them at the star point.

3 Test and Measurement

On first wake-up I started by measuring quiescent voltages. The voltages around Q_1, Q_2, and U_1 shown in table 1 seemed reasonable. Voltages around U_2 were all near ground except for the power supply pins. I then connected a signal generator to the input. Gain and headroom seemed reasonable with a sine wave input. I used a home-made notch filter set to measure total harmonic distortion and noise (THD+N) [5]. This is a weak test for harmonic distortion, since the equalization reduces harmonics in the output. It is a better test however for hum and ripple which are emphasized by the equalization. I found THD+N to be more than 70 decibels below the input at 100 hertz, 1 kilohertz, and 10 kilohertz with an output of 5 volts AC. Then I switched to a square wave signal and obtained the result seen in figure 4. The output traces show the effect of de-emphasis in my preamp, as expected.

Now the important part: measuring frequency response compared to the RIAA transfer function. Using a set of Python scripts I've been working on recently [6] I built a wave file of tone bursts at logarithmically spaced frequencies, with phase and amplitude modified as prescribed in the RIAA pre-emphasis transfer function. I recorded a playback of this stimulus file through the phono

Table 1: Left channel quiescent voltages in phono stage. Right channel was quite similar. Readings were taken with a digital multimeter, input and output remained open-circuit.

Part	**Pin**	**Voltage**
Q_1	E	-0.608
Q_1	B	-0.023
Q_1	C	12.75
Q_2	E	13.40
Q_2	B	12.75
Q_2	C	0.001
U_1	1	-1.378
U_1	2	-0.003
U_1	3	0.000
U_1	4	-13.61
U_1	8	13.40

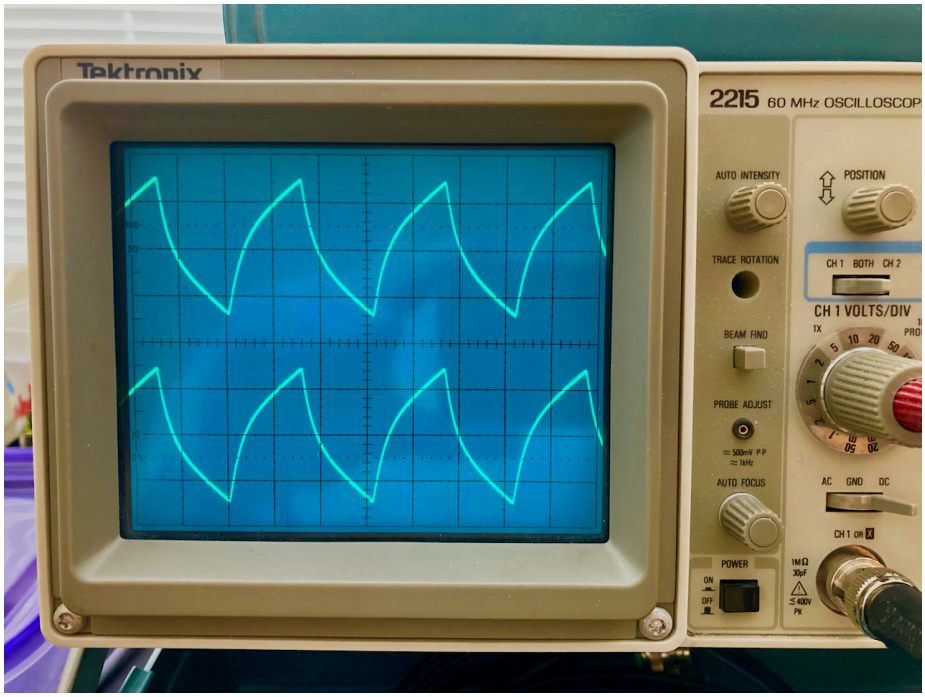


Figure 4: A quick look at how the phono stage responds to a square wave input shows good symmetry and no ringing. Output here is 14 V_{p-p} in both channels. Having an oscilloscope handy also lets us display a Lissajous pattern, to check cartridge tracking in loud passages.

preamp, and used another Python script to obtain the phase and amplitude of the preamp's output. Which precisely matched the computed transfer function of the servo bias amplifier. In other words, digital computation of the pre-emphasis curve in the Python script and analog computation of the de-emphasis curve in the preamp were the exact inverse of each other, within ± 0.1 decibels and ± 1.0 degree over the range 16 hertz ... 16 kilohertz. No trimming, and no binning, just solder in the parts and it works!

4 Listening Test

I'm old enough that I no longer rely on my ears to judge the merits of my designs. What's amazing to me is how much content and detail remains in the grooves of my modest record collection. After hours of focused listening I feel very comfortable with the JFET opamps in the UAF42s. My state-variable phono stage (figure 5) has delivered exactly the sort of meaningful enjoyment I was hoping for. Best wishes to everyone reading this, on your audio journey.

5 Special Thanks

Special thanks are due to the vendors who made this project possible:

- Tonläget HiFi, purveyor of the Linn Sondek LP12 transcription turntable and Lejonklou Slipsik 8 moving magnet phono stage.
- Akkelis Audio, purveyor of the Copland DAC215 headphone amplifier.
- Mouser Electronics, purveyor of all the bits and pieces needed to build the state-variable phono stage.



Figure 5: UAF42 phono stage in action. Surprisingly listenable for a first prototype. With no glaring deficiencies to work on, I'll just enjoy playing some records for now! Furniture by IKEA is quite stable and offers good support for the turntable.

Table 2: Bill of materials for the left channel. Note that R14, R16, C3, and C4 are internal to integrated circuit U2. Right channel is a duplicate except for a few parts shared across both channels including U1, R18, R19, D1, D2, C6, and C8. R13 is made of 316K and 2K in series. Enclosure, circuit board, and power supply are one of each. Incidental hardware and supplies not shown.

Designator	**Value**	**Description**	**Manufacturer**
R_1	22R	resistor	any
R_2	47K	precision film resistor	Vishay
R_3	3K	precision film resistor	Vishay
R_4	3K	precision film resistor	Vishay
R_5	27K	precision film resistor	Vishay
R_6	1K5	resistor	any
R_7	7K5	precision film resistor	Vishay
R_8	75K	precision film resistor	Vishay
R_9	10M	resistor	any
R_{10}	7K5	precision film resistor	Vishay
R_{11}	75K	precision film resistor	Vishay
R_{12}	7K5	precision film resistor	Vishay
R_{13}	318K	precision film resistor	Vishay
R_{14}	50K	integrated resistor	(internal to UAF42)
R_{15}	50K	precision film resistor	Vishay
R_{16}	50K	integrated resistor	(internal to UAF42)
R_{17}	47R	resistor	any
R_{18}	22R	resistor	any
R_{19}	22R	resistor	any
C_1	100P	mica capacitor	Knowles
C_2	470N	stacked film capacitor	Panasonic
C_3	1000P	integrated capacitor	(internal to UAF42)
C_4	1000P	integrated capacitor	(internal to UAF42)
C_5	100N	ceramic bypass capacitor	Kemet
C_6	4700U	electrolytic capacitor	Panasonic
C_7	100N	ceramic bypass capacitor	Kemet
C_8	4700U	electrolytic capacitor	Panasonic
D_1	1N4002	rectifier diode	Diotec
D_2	1N4002	rectifier diode	Diotec
Q_1	2N5089	NPN transistor	Onsemi
Q_2	2N5087	PNP transistor	Onsemi
U_1	TL082CP	dual JFET opamp	Texas Instr.
U_2	UAF42AP	universal active filter	Burr-Brown, Texas Instr.
	HAD15-0.4-A	linear power supply	SL Power
	1590DEBK	metal enclosure	Hammond Mfgr.
	8300SB1	circuit board	Twin Industries

References

[1] "RIAA Equalization," *Wikipedia*, 11 May 2026, `https://en.wikipedia.org/wiki/RIAA_equalization`.

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[3] D. Lancaster, *Active Filter Cookbook*, 2nd ed. Synergetics Press, Thatcher Arizona, 1995.

[4] M. Williamsen, "Using BodeS to Plot an RIAA Filter," *GitHub*, 11 May 2026, `https://github.com/springleik/BodeS/blob/master/RIAAFilterCoefficients.md`.

[5] M. Williamsen, "Notch-Filter Design," *Audio Electronics*, January 2000, pp. 10–17.

[6] M. Williamsen, "How to Obtain Results of audioXpress Paper," *GitHub*, 11 May 2026, `https://github.com/springleik/acBridge/blob/main/ObtainRIAAResults.md`.



Mark Williamsen was an intern in the lab at Shure Brothers during their 50th anniversary. Fifty years later he continues to enjoy vinyl audio. His more recent interests revolve around mixed domain signal processing and measurement. But when it all becomes too much, he finds comfort in traditional folk culture and music.